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Research paper

Anchoring behaviors and optimization of mechanical-bonding joint for FRP tendon

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A R T I C L E I N F O

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A B S T R A C T

Fiber-reinforced polymer (FRP) tendons have great potential in marine engineering due to their light weight, high strength, and corrosion resistance. However, anchor joints, as critical connections, are prone to sudden failure, which severely compromises structural durability. This study proposes a novel mechanical-bonding joint and develops a three-dimensional finite element model based on the Johnson–Cook model and cohesive zone model to investigate the mechanical response and failure process. The effects of key design parameters, including wedge angle, differential angle, joint length, and slotting length, on tensile capacity and compressive stress are systematically analyzed. A genetic algorithm–back propagation neural network is employed to establish a mechanical behavior prediction model and perform multi-objective optimization. Experimental results show that before optimization, the joint exhibited a tensile capacity of 300 kN with an anchorage efficiency of 63.1 %, and failure occurred in the clamping zone. After optimization (wedge angle 1.2°, differential angle 0.8°, joint length 150 mm, and inner pipe slotting length 125 mm), the tensile capacity increased to 452.4 kN, the anchorage efficiency improved to 95.2 %, and the failure mode shifted to central rod splitting. These results demonstrate the high reliability of the proposed optimization design method for high-strength mechanical-bonding joints.

1. Introduction

Fiber-reinforced polymer (FRP) tendon characterized by lightweight, high strength, and corrosion resistance possess significant potential for utilization (Zhao et al., 2020; Shi et al., 2016; Wang et al., 2025a) in maritime engineering applications including mooring lines, submarine cables and offshore wind turbine shown in Fig. 1 (Lu et al., 2020, 2024; Dong et al., 2018; Chen et al., 2025; Wang et al., 2025b; Zhang et al., 2023a, 2023b). One of the key components in ensuring the safety of FRP tendon systems is anchoring joint. Traditional adhesive anchoring is prone to aging and degradation of the adhesive layer in harsh marine environments of extreme pressure, high humidity and heavy salt spray (Li et al., 2023, 2024a; Akpinar and Aydin, 2014; Biscaia et al., 2016). In contrast, mechanical anchor has the advantages of superior strength, convenient construction and environmental durability, particular the sleeve-wedge anchors, which becomes a commonly connection scheme.

However, sleeve-wedged anchors are prone to extreme stress concentrations in the anchoring zone, leading to fiber crushing or matrix cracking, severely limiting anchoring efficiency. Therefore, mechanical behavior and structural design of high-strength anchoring joints must be thoroughly examined to increase the reliability of FRP tendon (Hammerl et al., 2024).

The main challenge in sleeve-wedge anchoring systems for FRP tendons lies in the effective transfer of the high tensile loads from the composite tendon to the anchor, and then to the structure without damage or failure (Mei et al., 2023). Experts and scholars from both domestic and foreign countries have conducted a large number of researches. For the anchoring mechanism, Schmidt et al. (Schmidt et al., 2012; Li et al., 2024b). pointed out that weak performance of sleeve-wedge anchoring systems is often due to tendon brittleness, low strength perpendicular to the fiber direction, and insufficient stress transfer at the interface, leading to excessive principal stresses, local crushing, and interfacial slippage. For the optimization of anchoring scheme, Feng et al. (2019) proposed a segmented variable stiffness anchoring structure for FRP tendons, studying failure modes, anchoring efficiency, stress and displacement in the anchor zone. Optimized designs can achieve high anchoring efficiency of up to 91 %. For the influencing factors of anchoring performance, Xie et al. (2019) proposed a universal theoretical model to predict stress distributions in sleeve-wedge anchors for carbon fiber-reinforced polymers (CFRP)

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tendons. Slip resistance increases with conical degree, but stress concentration increases as well. They suggested a conical degree between 3° and 4° for anchors with an anchor length of no less than 200 mm. Ye et al. (2019) investigated the effects of wedge thickness, length of wedge corner cut, roughness of contact surface, and pretension load on anchoring performance. Their findings indicated that both sliding and interlaminar debonding, rather than tensile rupture, often determine the maximum anchoring force, especially for compact anchorages. Li et al. (Huang and Li, 2023; Zhang et al., 2022; Chang and Chen, 2019; Wang et al., 2023) studied the influence of geometric parameters of wedge-shaped anchors (including the size and material of the wedge, the friction coefficient, and the preload) on the stress distribution of basalt fiber reinforced polymers (BFRP) plates. The study shows that the thickness and the friction coefficient significantly affect the peak compressive stress, while the elastic modulus is the main factor affecting the peak shear stress. In summary, scholars both at home and abroad have conducted extensive research on the clamping performance and influencing factors of FRP wedge-type anchorages. The key challenge lies in how to significantly reduce the lateral compression damage to the FRP reinforcement during anchorage through structural design, in order to avoid brittle failures such as fiber crushing or delamination. Currently, the adhesive-mechanical composite type anchorages are recognized as the most promising solution. However, the coupling mechanism of adhesion and mechanical clamping in this system still needs to be further revealed. In particular, the load distribution law, interface stress-strain evolution path, and the collaborative design theory of the anchoring system considering multi-parameter coupling effects have not been fully developed (Wang et al., 2015), which restricts the further optimization and application of high-performance anchoring structures. At the same time, the mechanical-bonding anchor system is often evaluated only on the tensile strength, neglecting the radial compressive stress on the FRP tendon. This one-sided evaluation system fails to reveal the internal damage accumulation due to lateral compression, making it difficult to predict and prevent premature failure modes like low-stress radial crushing or brittle pull-out (Xu et al., 2019; Reda et al., 2016).

In this paper, a mechanical-bonding anchor scheme was presented in this study employing a high-precision three-dimensional finite element analysis model that took into account the cohesion zone model (CZM) and Johnson-Cook (J-C) constitutive model, which accurately depicts complicated mechanical processes and progressive failure under pulling stress. An effective anchoring system was firstly created using a dual-objective optimization approach that combined pulling load and compressive stress to identify parameter combinations, creating an efficient anchoring system.

2. Theoretical model

2.1. Anchoring structure

The mechanical-bonding anchoring joint shown in Fig. 2 uses self-locking friction on the inclined surface between inner-wedged tubes and outer sleeve to convert lateral pressures into compressive forces. The clamping force independently increases with the load under self-locking conditions, creating a robust mechanical restraint. The interface between FRP and steel is fully saturated with adhesive. Under the combined action of mechanical and chemical forces, the overall stiffness of the joint is significantly improved. Moreover, the outer sleeve is made of high-strength alloy steel 40Cr, while the inner-wedged tubes are fabricated from 35CrMo, which features multiple hollow conical ring segments.

The principle of the mechanical-bonding anchoring joint can be categorized into three stages.

(a) Tightening stage: A radial compressive force is generated by the relative movement between the outer sleeve and inner-wedged tubes, resulting in the mechanical fastening of the FRP tendon.
(b) Self-locking stage: In the absence of external pressure, the inner tube shows a tendency to pull back due to elastic recovery, which creates a self-locking reverse frictional force.
(c) Pulling out stage: When tensile force applied on the FRP tendon, this load is transferred to the anchorage system through friction and adhesion at the interface, resulting to a dynamic load-bearing mechanism of the anchoring joint.

2.2. Theoretical behavior analysis

The force analysis for tightening stage, self-locking stage and pulling out stage is shown in Fig. 3.

(1) Tightening stage

Take one inner-wedged tube as the isolated object. Utilizing the concept of symmetry, the spatial force is simplified to a planar force. The relationship of forces at pre-tightening stage in the vertical and horizontal directions is (Jiang and Fang, 2008):

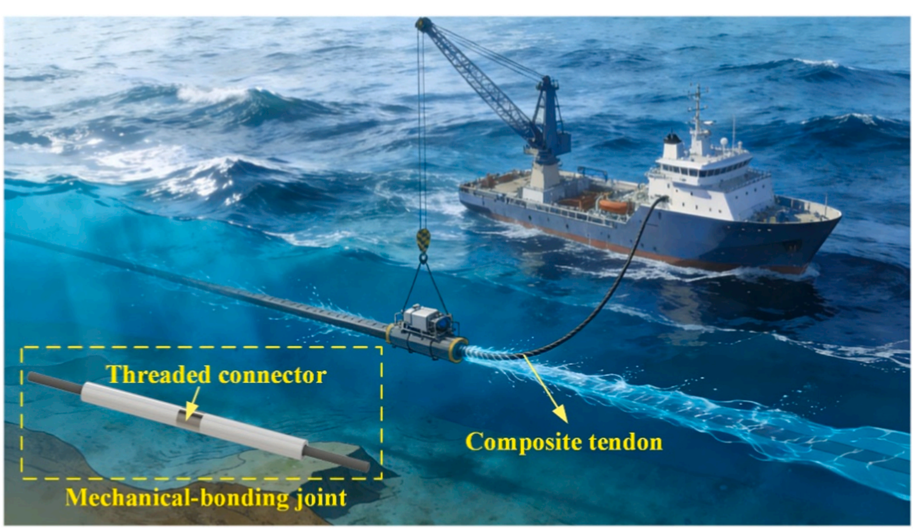


Fig. 1. Mechanical anchoring joint in marine engineering.

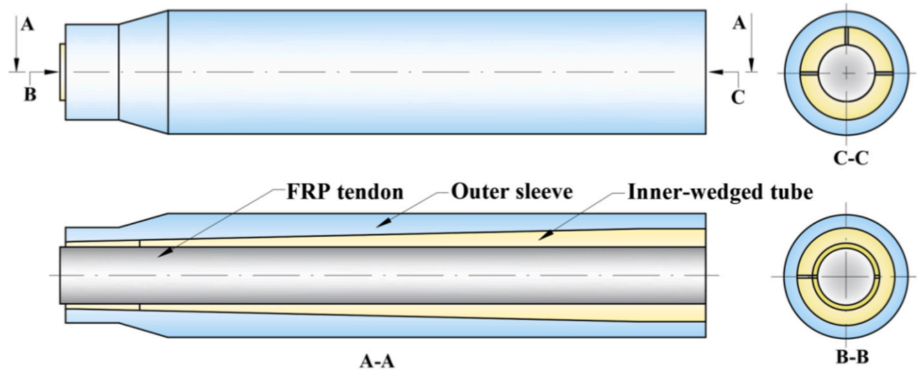


Fig. 2. The scheme of mechanical-bonding anchoring joint.

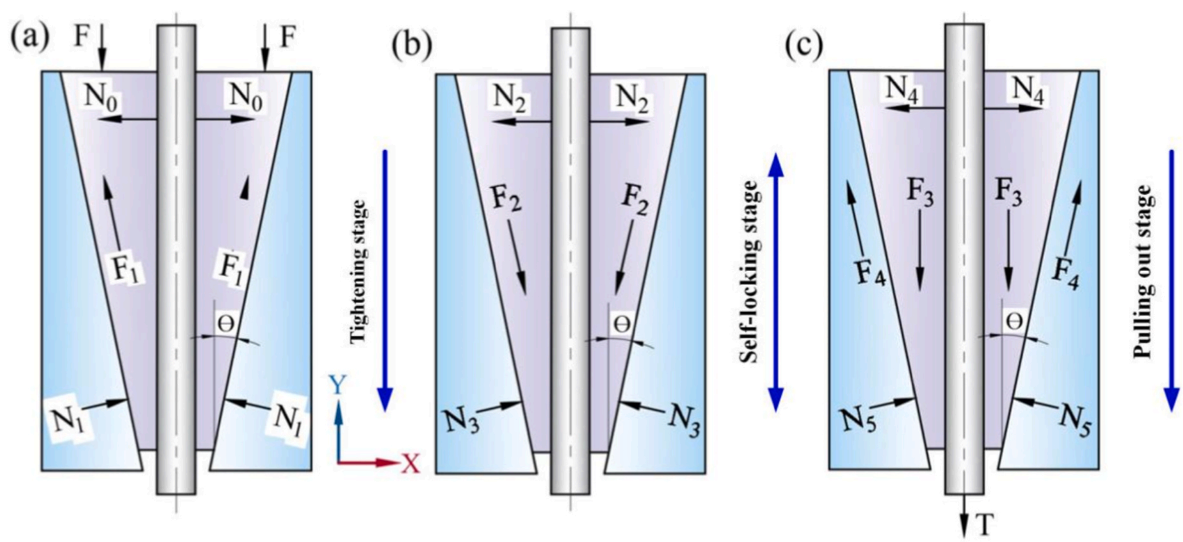


Fig. 3. Working principle of mechanical-bonding anchoring joint.

$$\begin{cases} \sum F_x = 0;\ N_0 + F_1 \sin\theta - N_1 \cos\theta = 0 \\ \sum F_y = 0;\ F - F_1 \cos\theta - N_1 \sin\theta = 0 \\ F_1 = \mu N_1, \mu = \tan\alpha \end{cases} \tag{1}$$

where F represents the preload force of the inner-wedged tube; N_0 represents the normal force between the tube and tendon; F_1 represents the frictional force between the tube and sleeve; θ is the inclination angle of the tube and the sleeve; α is the friction angle between the tube and sleeve.

Thus, the pre-tightening force of the tube is obtained as

$$F = N_0 \frac{\tan\alpha + \tan\theta}{1 - \tan\alpha \tan\theta} = N_0 \tan(\alpha + \theta) \tag{2}$$

(2) Self-locking stage

The relationship of forces at unloading stage in the vertical and horizontal directions is:

$$\begin{cases} \sum F_x = 0;\ N_2 - F_2 \sin\theta - N_3 \cos\theta = 0 \\ \sum F_y = 0;\ F_2 \cos\theta - N_3 \sin\theta = 0 \\ F_2 = \mu N_3, \mu = \tan\alpha \end{cases} \tag{3}$$

For the tube to remain locked within the sleeve, the subsequent conditions must be satisfied:

$$\tan\alpha \geq \tan\theta \tag{4}$$

To prevent the tube from rebounding to the sleeve after pre-tightening and subsequent unloading, the cone angle of the tube must be less than or equal to the friction angle $\alpha \geq \theta$.

(3) Pulling out stage

The relationship of forces at operating stage in the vertical and horizontal directions is:

$$\begin{cases} \sum F_x = 0;\ N_4 + F_4 \sin\theta - N_5 \cos\theta = 0 \\ \sum F_y = 0;\ F_3 - F_4 \cos\theta - N_5 \sin\theta = 0 \\ F_4 = \mu N_5, \mu = \tan\alpha \end{cases} \tag{5}$$

From the equilibrium relationship of the horizontal force in the tube, it can be known that:

$$N_4 = \frac{N_5 \cos(\alpha + \theta)}{\cos\alpha} \tag{6}$$

From the equilibrium relationship of the vertical force in the tube, it can be known that

$$F_3 = \frac{N_5 \sin(\alpha + \theta)}{\cos\alpha} \tag{7}$$

Substituting equation (7) into equation (6) results

$$F_3 = N_4 \tan(\alpha + \theta) \tag{8}$$

The conditions for ensuring that the joint is securely anchored without any slippage are as follows:

$$F_3 \leq \tan \kappa N_4 \tag{9}$$

where κ represents the friction angle between the tube and tendon. Substituting equation (9) into equation (8), one obtains

$$\tan(\alpha + \theta) \leq \tan \kappa \tag{10}$$

To guarantee that the tendon is securely secured without slippage, the cone angle and the friction angle between the tube and sleeve must be less than or equal to the friction angle between the tendon and the tube, $\alpha + \theta \leq \kappa$.

3. Finite element model

3.1. Constitutive model

For mechanical behavior, the Johnson-Cook model is used to describe the strength limitations and failure mechanisms in metallic materials subject to extreme strains, elevated strain rates, and high temperatures. The flow stress or yield stress is formed as

$$\sigma = (A + B\varepsilon^n)\left(1 + C\ln\frac{\dot{\varepsilon}}{\dot{\varepsilon}_0}\right)(1 - T^{*m}) \tag{11}$$

$$T^* = \frac{T - T_r}{T_m - T_r} \tag{12}$$

where σ represents flow stress; ε, $\dot{\varepsilon}$, and $\dot{\varepsilon}_0$ represent equivalent plastic strain, loading strain rate, and reference strain rate, respectively; T, T_m, and T_r represent experimental temperature, material melting temperature, and ambient temperature, respectively; and A, B, C, m and n represent the material parameters and can be determined by material experiment. As the main component of the workpiece material is ferrite, so the constitutive model parameters of ferrite were used to set as the parameters of the workpiece. And the Johnson-Cook constitutive model parameters of ferrite are shown in Table 1 (Wang et al., 2025c).

At the same time, the Johnson-Cook damage criterion needs to be added to the model, which considers the equivalent failure strain of strain rate, compressive stress state, and temperature in the deformation area. Under this failure criterion, the equivalent of the element integration point of workpiece plastic strain is used to define whether it fails or not. Its equation is as follows:

$$D = \int \frac{d\bar{\varepsilon}_p}{\bar{\varepsilon}_p^f} \tag{13}$$

where $d\bar{\varepsilon}_p$ is the equivalent plastic strain increment of the integral element, and $\bar{\varepsilon}_p^f$ is the critical equivalent plastic failure strain.

When the damage parameter D exceeds 1, the integral element is judged to be deleted after failure.

The expression of $\bar{\varepsilon}_p^f$ is shown as follows:

$$\bar{\varepsilon}_p^f = \left[d_1 + d_2 \exp\left(d_3\frac{\sigma_p}{\sigma_m}\right)\right] \times \left[1 + d_4 \ln\left(\frac{\dot{\varepsilon}_p}{\dot{\varepsilon}_0}\right)\right] \times \left[1 + d_5\frac{T - T_r}{T_m - T_r}\right] \tag{14}$$

where $\bar{\varepsilon}_p^f$ is the critical equivalent plastic failure strain, σ_p is the compressive stress, σ_m is the Mises stress, $\dot{\varepsilon}_p$ is the strain rate, and $\dot{\varepsilon}_0$ is the reference strain rate. The parameters of the Johnson-Cook damage model of ferrite are shown in Table 2.

For bonding behavior, according to the actual stress of the engineering structure, I, II, III crack or their combinations may appear in the bonded interface.

Table 1
Johnson-Cook constitutive model parameters of ferrite.

A/MPa	B/MPa	C	m	n	T_m/°C
190	600	0.012	0.937	0.18	1800

Table 2
Johnson-Cook damage model parameters of ferrite.

d_1	d_2	d_3	d_4	d_5
0.05	4.22	−2.73	0.0018	0.55

In the linear elastic stage of the bilinear mixed-mode cohesion model, the damage is assumed to initiate when a quadratic interaction function involving the nominal stress ratios reaches a value of 1, and its basic formula (1) is listed as follows (Zhang et al., 2023c):

$$\left(\frac{t_n}{t_n^0}\right)^2 + \left(\frac{t_s}{t_s^0}\right)^2 + \left(\frac{t_t}{t_t^0}\right)^2 = 1 \tag{15}$$

where t_n, t_s and t_t represent the normal and the shear tractions, t_n^0, t_s^0 and t_t^0 represent the peak values of the nominal stress when the deformation is either purely normal to the interface or purely in the first or the second shear direction.

In the stage of damage, the damage evolution criterion proposed by Benzeggagh and Kenane (B-K) is used as the damage propagation criterion in the mixed mode in equation (2) as follows:

$$G_{IC} + (G_{IIC} - G_{IC})\left(\frac{G_{shear}}{G_T}\right)^\eta = G_C \tag{16}$$

Among them,

$$G_{shear} = G_{II} + G_{III}, G_T = G_I + G_{II} + G_{III} \tag{17}$$

where G_C is the failure equivalent critical fracture energy; and G_I, G_{II},G_{III} are the fracture energy in I,II,III crack type of cohesive mode; G_{IC}、 G_{IIC} are the critical fracture energy in each direction; and η is the damage factor determined by test. The material parameters of the interface are listed in Table 3 (Zhang et al., 2024).

3.2. Numerical simulation

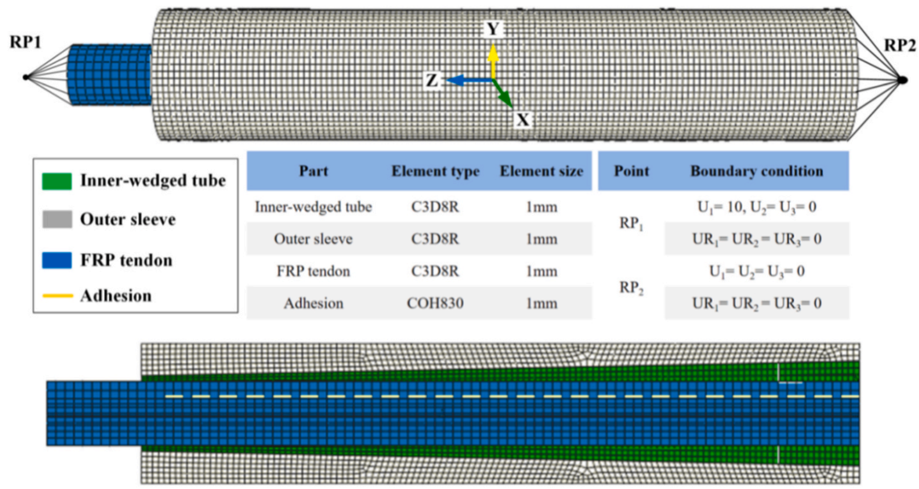
A three-dimensional simulation model of the mechanical-bonding anchoring joint was developed in Abaqus, including FRP tendons, inner-wedged tubes, and outer sleeve as shown in Fig. 4. Among them, the wedge angle defined 1.3°, the differential angle was 0°, the joint length was 125 mm, and the slotting length was 90 mm. The sweeping meshing technology was utilized along with hourglass control and the neutral axis algorithm to enhance computational stability. In the setting of constraint conditions, the right end of the outer tube was defined as fully fastened. An axial displacement load was applied in the Z direction to the left end face of the FRP tendon. The interface between the inner tube and the FRP tendon was characterized as sliding friction, with a friction coefficient established at 0.05.

3.3. Load-displacement curve

The numerical load-displacement curve of the pull-out test for mechanical-bonded joint is shown in Fig. 5. In the elastic deformation stage, as the displacement reached 3.5 mm, the pull-out force rose linearly to 841.9 kN. As the displacement increased from 3.5 mm to 5.0 mm, the joint entered the yield stage, leading to a reduced rate of

Table 3
Interfacial material parameters.

Parameters	Value
T_n/MPa	20
T_s、 T_t/MPa	25
G_{IC}/N•mm	0.25
G_{IIC}/N•mm	0.7



Part	Element type	Element size	Point	Boundary condition
Inner-wedged tube	C3D8R	1mm	RP_1	$U_1 = 10$, $U_2 = U_3 = 0$
Outer sleeve	C3D8R	1mm		$UR_1 = UR_2 = UR_3 = 0$
FRP tendon	C3D8R	1mm	RP_2	$U_1 = U_2 = U_3 = 0$
Adhesion	COH830	1mm		$UR_1 = UR_2 = UR_3 = 0$

Fig. 4. The simulation model of the mechanical-bonding anchoring joint.

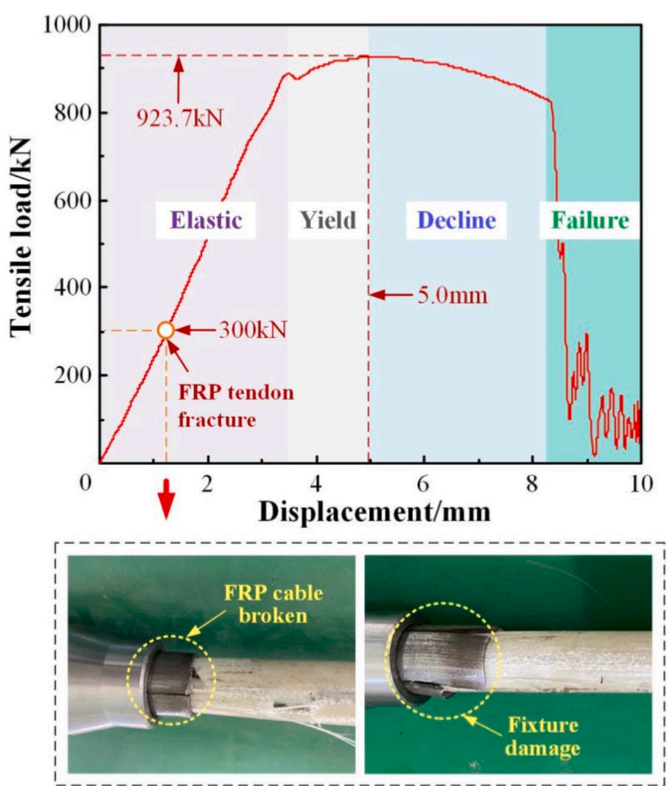


Fig. 5. Load-displacement curve of mechanical-bonded anchor joint.

increase in pull-out load, which ultimately reaches a peak load of 923.7 kN. Subsequently, the joint entered a stable decline stage, during which the pull-out load steadily decreased to 785.2 kN, followed by a fast failure stage characterized by a sudden reduction in the pull-out load.

However, during working conditions, the joint suffered radial crushing fracture when pulling force reached 300 kN with anchoring efficiency of 63.1 %. This is because the FRP tendon has a substantially lower transverse strength than axial tensile strength. Due to radial pressure in the anchoring region, FRP tendons experienced crushing damage before completely utilizing tensile strength. Thus, increasing anchoring system stress distribution to prevent transverse compression damage can increase bearing capacity of anchor joint. This provides a foundation for emphasizes the importance of controlling radial stress concentration to improve FRP anchoring system.

The stress distribution of the mechanical-bonding anchor joint is illustrated in Fig. 6 when the tensile displacement was 10 mm. The stress concentration at the interface of the inner-wedged tube and FRP tendon was significant. The peak compressive stress attained was 608.1 MPa, mostly concentrated in the slotting area near the fixed end, and gradually diminished in the axial direction towards the loading end. The shear stress distribution exhibited the same decreasing trend, with a maximum shear stress of 349.7 MPa attained.

Meanwhile, the highest compressive stress at the interface between the inner-wedged tube and outer sleeve is 492.6 MPa. A significant range of reverse compressive stress is observed in the one-third region adjacent to the fixed end of the outer sleeve. That is because the joint reached the quick failure stage, leading to a strength reduction in the inner tube with plastic deformation or localized failure first exhibiting roughly at the one-third position near the fixed end. The inner tube inadequately and inconsistently passed the compressive force, resulting in a stress turnaround and an uneven shear stress distribution.

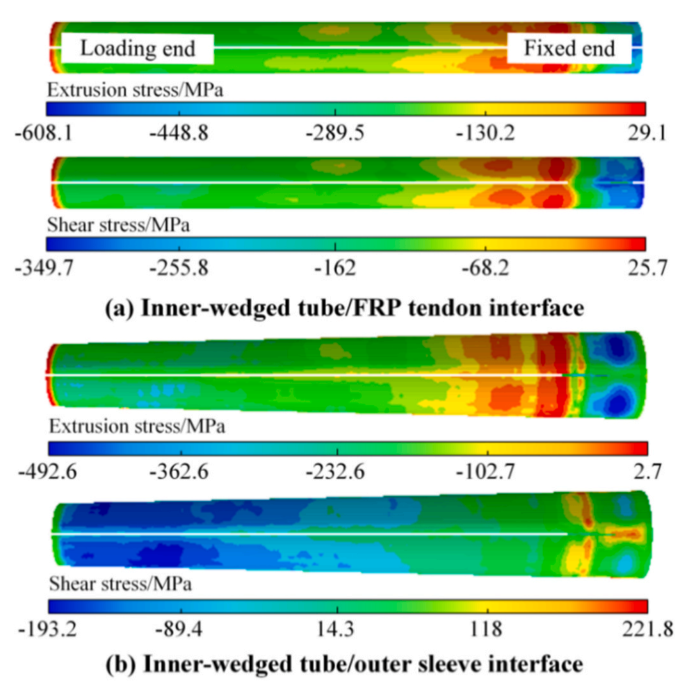


Fig. 6. Stress distribution of the mechanical-bonding anchor joint under 10 mm tensile displacement.

4. Influence factor analysis and optimum

To explore the factors influencing the strength of mechanical-bonding anchor joints, the anchoring performance was evaluated for different wedge angles, differential angles, joint lengths, and slotting lengths, as shown in Fig. 7.

4.1. Wedge angle

A simulation was conducted to examine the effect of the wedge angle on the anchoring strength of the mechanical-bonding anchor joint, featuring a differential angle β of 0° between the inner-wedged tube and outer sleeve, a joint length l of 125 mm, and a slotting length t of 90 mm. The fluctuation of anchoring strength was examined at wedge angles α of 0.9°, 1.1°, 1.3°, 1.5°, and 1.7°, as illustrated in Fig. 8.

As the wedge angle α grew from 0.9° to 1.7°, the ultimate tensile stress of the joint rose from 853.9 kN to 983.5 kN, showing a 15.2 % increase; however, the initial damage displacement decreased from 5.3 mm to 4.8 mm. This alteration signified that an increase in the angle reduced the ductile limit while enhancing the anchoring strength. With a tensile displacement of 2 mm, the compressive stress at the contact surface of the inner-wedged tube/FRP tendon showed an obvious upward variation in the axial direction. At an angle α of 0.9°, the compressive stress rapidly increased to a maximum of 380.5 MPa within the range of 0 mm–112 mm, subsequently declining sharply; as the wedge angle α gradually rose to 1.7°, the peak stress progressively increased to 413.8 MPa, 432.1 MPa, 439.8 MPa, and 437.2 MPa, reflecting an overall increase of approximately 14 %.

As the wedge angle grows, the radial embedding degree between the inner-wedged tube and outer sleeve intensifies. In the case of a short angle, inadequate radial feed leads to lowered normal compressive stress, which results in stress variations at the loading end. As the wedge angle increases, the radial feed increases, and a more adequate and uniform normal compression is established, thereby augmenting the clamping strength of the FRP tendon, ultimately resulting in a significant rise in the ultimate tensile load. Nevertheless, an extremely wide wedge angle may induce radial crushing of the FRP tendon. The wedge angle is a critical design element for regulating the mechanical properties and failure mode of the joint.

4.2. Differential angle

To examine the effect of differential angle on the anchoring strength, a mechanical-bonding anchor joint model was created with a wedge angle α of 1.3°, a joint length l of 125 mm, and a slotting length t of 90 mm. The change of anchoring strength was examined at differential angle β of 0°, 0.5°, 1.0°, 1.5°, and 2°, as seen in Fig. 9.

The variation in differential angle between the inner-wedged tube and outer sleeve significantly impacted the failure mode and mechanical properties of the mechanical-bonding anchor joint. At a differential angle β of 0°, the joint exhibited ductile failure behaviors. As the differential angle β grew to 0.5°, the failure mode changed to brittle fracture, and the maximum pull-out load reached 1101.7 kN. An increase in the differential angle β to 2° resulted in a substantial drop in the ultimate pull-out load to 425.4 kN, reflecting a reduction of 61.4 %.

With a pull-out displacement of 2 mm, the increase in differential angle β resulted in a major decrease in the compressive stress applied by the inner tube on the FRP tendon, declining from 439.8 MPa to 170.0 MPa with a reduction of 61.3 %. Furthermore, the stress distribution in the axial direction became increasingly uniform.

The fundamental mechanism can be attributed to the following: At 0° differential angle, complete contact between the inner-wedged tube and outer sleeve allowed stress transfer by uniform mechanical interlocking, resulting in plastic deformation. The insertion of a differential angle influenced the geometric parameters of the contact interface. While it slightly decreased the effective contact area and lowered local mechanical interlocking, it significantly enhanced the distribution of radial pressure and alleviated stress concentration, thus augmenting the pull-out load. However, as the differential angle increased, the contact area reduced sharply, leading to a reduction in the mechanical interlocking effect, which in turn resulted in decreased radial pressure and impaired load-transfer capability. As a result, the bearing performance of the joint weakened. Thus, handling the differential angle required balancing connection strength and optimum stress distribution to improve interface performance.

4.3. Joint length

A mechanical-bonding anchor joint model was developed to investigate the influence of joint length on anchoring strength, featuring a wedge angle α of 1.3°, a differential angle β of 0° and a slotting length t of 90 mm. The change in joint length l was analyzed for 120 mm, 135 mm, 150 mm, 165 mm, and 180 mm, as illustrated in Fig. 10.

As the joint length l of the mechanical-bonding anchor joint increased from 120 mm to 180 mm, the ultimate pulling load climbed from 902.3 kN to 998.3 kN, reflecting an approximate increase of 10.6 %. The initial failure displacement sequentially increased from 5.25 mm to 5.70 mm, hence showing an enhancement in fracture toughness. At the same time, the maximum compressive stress between the inner-wedged tube and outer sleeve reduced from 439.8 MPa to 370 MPa (a decrease of 15.7 %), resulting in a more uniform stress distribution in the axial direction.

The improvement in performance is mainly due to the greater effective contact area resulting from the longer joint length. The load was conveyed via an increased number of mechanical interlocking points, which markedly lowers stress concentration, enhances the distribution of radial pressure, and alleviates stress concentration. As a result, the gripping strength of the joint was enhanced by about 3 %.

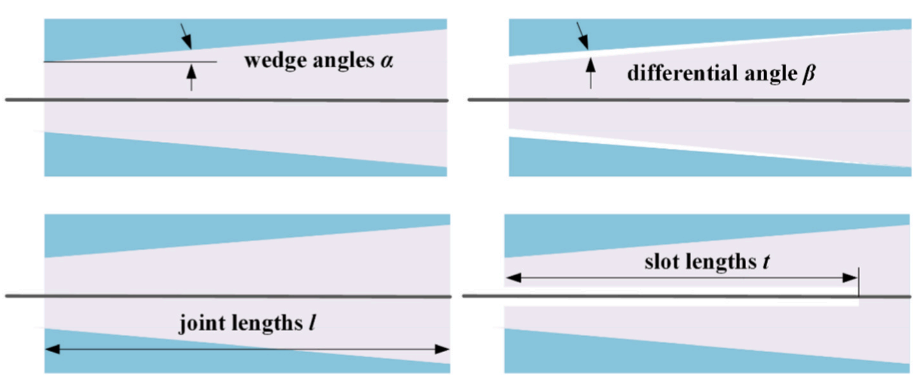


Fig. 7. Different design parameters of mechanical-bonding anchor joint.

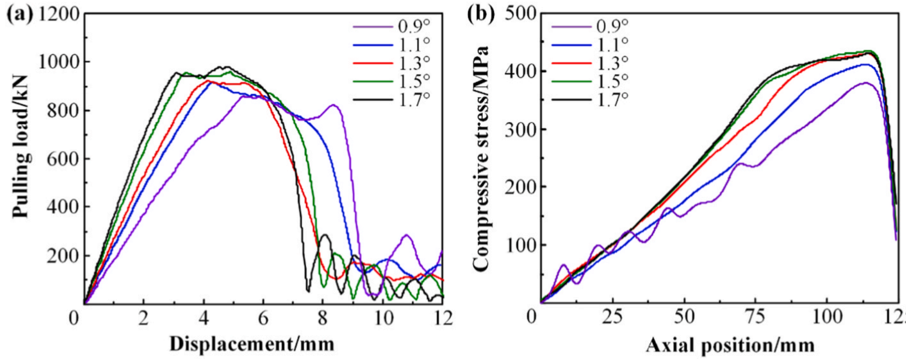


Fig. 8. The change of mechanical property of mechanical-bonding anchor joint at various wedge angles with (a) load-displacement curve and (b) compressive stress.

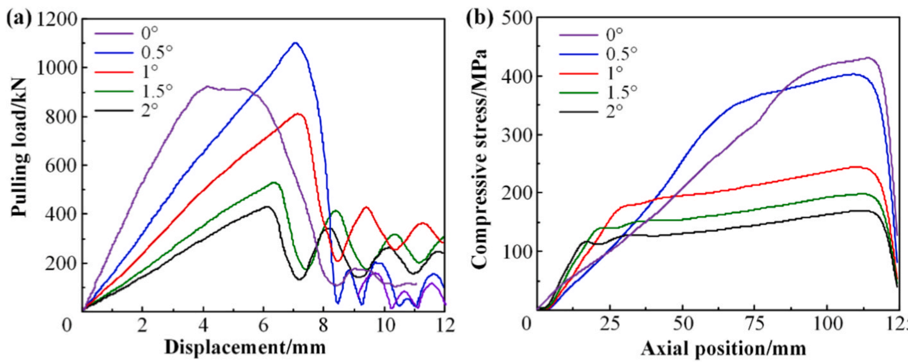


Fig. 9. The change of mechanical property of mechanical-bonding anchor joint at various differential angle with (a) load-displacement curve and (b) compressive stress.

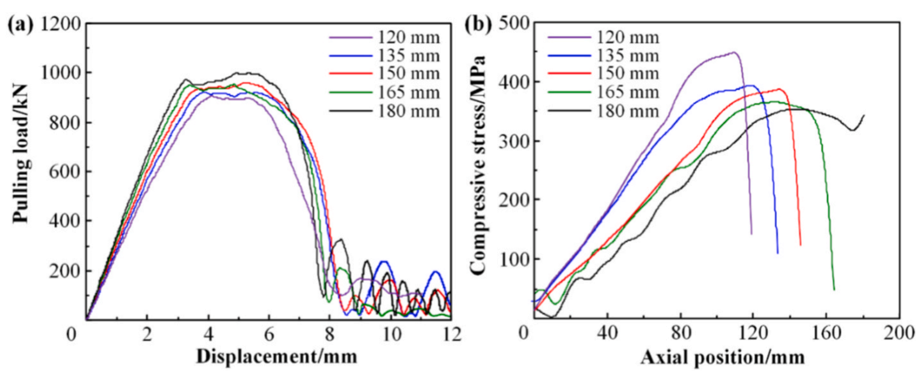


Fig. 10. The change of mechanical property of mechanical-bonding anchor joint at various joint length with (a) load-displacement curve and (b) compressive stress.

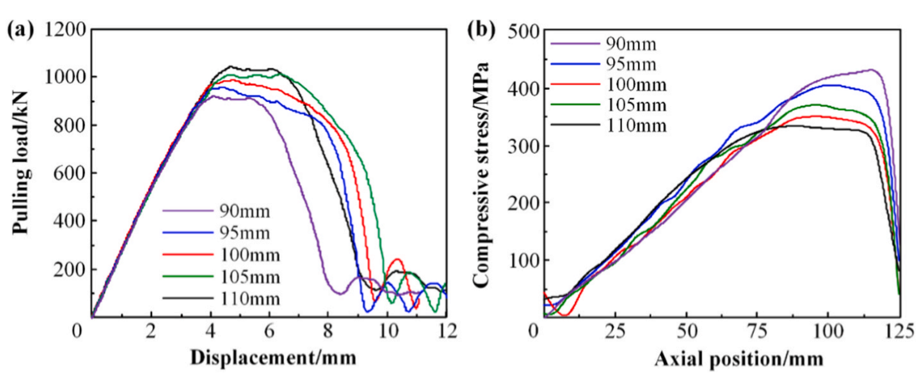


Fig. 11. The change of mechanical property of mechanical-bonding anchor joint at various slotting length with (a) load-displacement curve and (b) compressive stress.

4.4. Slotting length

A mechanical-bonding anchor joint model was created to examine the effect of the slotting length of the inner-wedged tube on anchoring strength, characterized by a wedge angle of $\boldsymbol{\alpha}$ 1.3°, a differential angle $\boldsymbol{\beta}$ of 0°, and a joint length $\boldsymbol{l}$ of 125 mm. The change in slotting length $\boldsymbol{t}$ was examined for 90 mm, 95 mm, 100 mm, 105 mm, and 110 mm, as shown in Fig. 11.

As the slotting length $\boldsymbol{t}$ of the inner-wedged tube increased from 90 mm to 110 mm, the ultimate tensile stress of the mechanical-bonding anchor joint increased from 923.7 kN to 1037.2 kN, indicating an improvement of 13.8 %. The displacement at initial failure rose from 4.0 mm to 4.7 mm. Simultaneously, the maximum compressive stress at the interface of the inner tube and the FRP tendon diminished from 445.9 MPa to 324.5 MPa, reflecting a decrease of 27.2 %.

This phenomenon is due to the lowered stiffness of the inner-wedged tube resulting from the increased slotting length. The reduced stiffness allowed the tube to better adhere to the FRP tendon under compression stress. The effective contact area was larger, promoting a more uniform distribution of the transferred load. The radial compressive stress concentrated on the FRP tendon was significantly decreased, improving the tensile strength of the joint. Overall, adjusting joint stiffness improves interface stress distribution, providing a theoretical framework for designing high-performance composite connection systems.

4.5. Optimum structural design

The Genetic Algorithm-Back Propagation (GA-BP) algorithm is a hybrid optimization model that combines the global search capacity of the GA with the local fine-tuning strength of the BP Neural Network. Within this framework, the GA initially conducts a global random search in the solution space utilizing evolution operations—selection, crossover, and mutation—to establish a high-quality initial population of neural network weights and thresholds. This method reduces drawbacks associated with conventional BP networks, including sensitivity to initial parameters and a tendency to converge to local minima. Subsequently, the BP algorithm optimizes initial parameters through the gradient descent method for accurate modification, reducing the prediction error and thus improving both convergence speed and predictive accuracy of the model.

This study employs the GA-BP algorithm to predict the mechanical performance of mechanical-bonding anchor joints. The model incorporates fundamental structural parameters of the joint as inputs, comprising the wedge angle, differential angle, joint length, and slotting length, which correspond to four neurons in the input layer. The output layer consists of a single neuron corresponding to the pull-out load, with radial pressure as a separate output. The ideal neural network architecture is 4–20–16–1. The network training parameters were defined as follows: a maximum of 1000 epochs, an initial learning rate of 0.1, a momentum coefficient of 0.3, and a target mean squared error (MSE) of 0.0001. The orthogonal experimental design technique selected a uniformly distributed and representative training sample from the comprehensive experiments, as shown in Table 4.

The genetic algorithm for mechanical-bonding anchor joint has a coding range of $[-1, 1]$. The number of input layer nodes in the network is 4, the number of hidden layer nodes is 6, and the number of output layer nodes is 1. The number of iterations is set to 200, the population size is 100, the crossover probability is 0.7, and the mutation probability is 0.3. The parameters of the genetic algorithm are shown in Table 5.

The GA-BP neural network model demonstrated the most favorable fitting effect within the designated parameter range shown in Fig. 12. The training dataset fitting R value of pulling load and compressive stress was 0.949 and 0.842, respectively. The GA-BP model exhibited exceptional fitment performance. Simultaneously, the training samples were employed to develop a prediction model of the mechanical-bonding anchor joint. The maximal error between the output results of the verification samples and the prediction model was 12.2 % and 16.5 %, respectively. The results indicate that the GA-BP neural network exhibits a strong capacity for generalization in the context of mechanical-bonding anchor joint under a variety of structural parameters.

Table 4
Sample set of mechanical-adhesive coupling joints.

Number	X_1	X_2	X_3	X_4	X_5	X_6
1	0.9	0	230	180	895.9	484.2
2	0.9	0.5	240	190	762.7	466.1
3	0.9	1	250	200	500.4	416.5
4	0.9	1.5	260	210	325.4	354.5
5	0.9	2	270	220	211.8	298.5
6	1.1	0	240	200	871.9	449.5
7	1.1	0.5	250	210	696.8	368.5
8	1.1	1	260	220	461.2	387.5
9	1.1	1.5	270	180	347.9	335.5
10	1.1	2	230	190	263.7	246.5
11	1.3	0	250	220	857.6	421.3
12	1.3	0.5	260	180	743.2	385.5
13	1.3	1	270	190	468.0	275.5
14	1.3	1.5	230	200	310.5	235.5
15	1.3	2	240	210	263.2	202.5
16	1.5	0	260	190	873.1	400.5
17	1.5	0.5	270	200	712.5	256.5
18	1.5	1	230	210	524.6	370.5
19	1.5	1.5	240	220	375.5	210.5
20	1.5	2	250	180	270.9	158.5
21	1.7	0	270	210	903.3	298.5
22	1.7	0.5	230	220	745.5	247.5
23	1.7	1	240	180	439.7	159.4
24	1.7	1.5	250	190	286.3	125.5
25	1.7	2	260	200	197.1	78.5

Table 5
Genetic algorithm for setting parameters.

Parameter	Value
Training algorithm	Levenberg-Marquardt
Transfer function	Hyperbolic tangent S-shaped curve
Network learning rate	0.005
Maximum number of network training steps	100000

Based on the prediction models of the clamping strength and compressive stress under different design parameters, a multi-objective optimization model for mechanical-bonding anchor joints was constructed. The trained GA-BP neural network was used as the fitness function of the NSGA algorithm. The population size was set to 80, the number of iterations was 3000, the crossover probability was 0.4, and the mutation probability was 0.4. With the pull-out load T and radial pressure P as the optimization objectives, the optimization objective can be expressed as:

$$\begin{cases} y_1 = \min(-T) \\ y_2 = \min(-P) \end{cases} \tag{18}$$

Optimizing the variables can be expressed as:

$$\begin{cases} 0.8 \le X_1 \le 1.2 \\ 0 \le X_2 \le 2 \\ 100 \le X_3 \le 300 \\ 80 \le X_4 \le 280 \end{cases} \tag{19}$$

The structural parameters that met the peak pulling load and minimum compressive stress were selected as the final optimization parameters. The optimized wedge angle was 1.2°, the differential angle was 0.8°, the joint length was 150 mm, and the inner pipe slotting length was 125 mm.

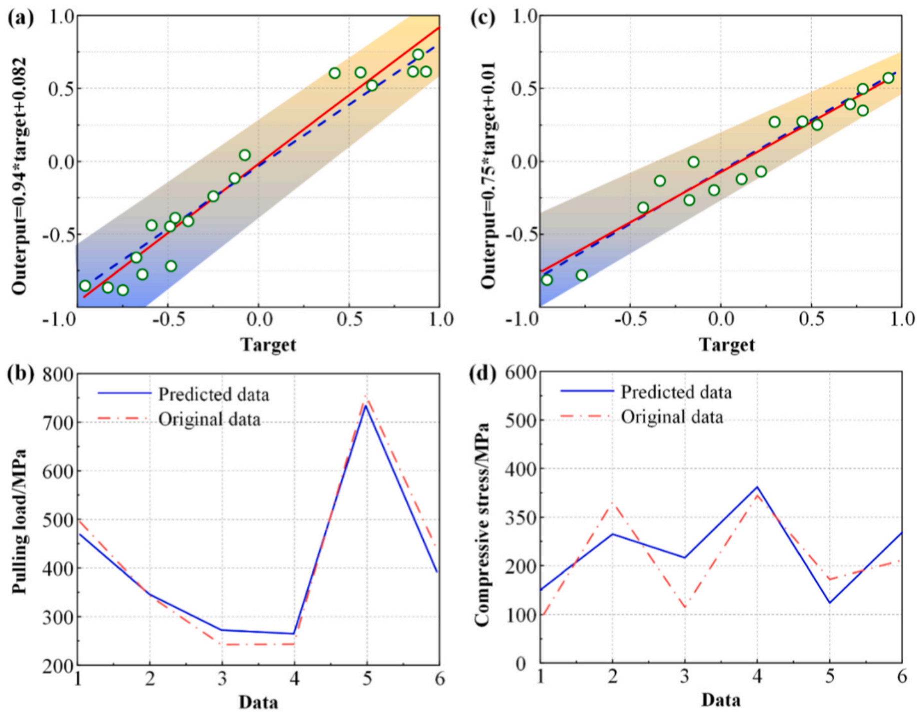


Fig. 12. Fitness curve of the optimum individual for the design parameters of mechanical-bonding anchor joint.

5. Experimental validation

5.1. Experimental design

FRP tendon produced by Shengli Xinda Composite Co., LTD. are manufactured using the pultrusion process. The raw materials used include carbon fiber, glass fiber and bisphenol A type epoxy resin in which the fiber volume fraction is approximately 70 %. Prior to the test, exert a force on the ends of the inner tube to enter the FRP tendon into the outer sleeve, which has been pre-filled with epoxy resin. Use the

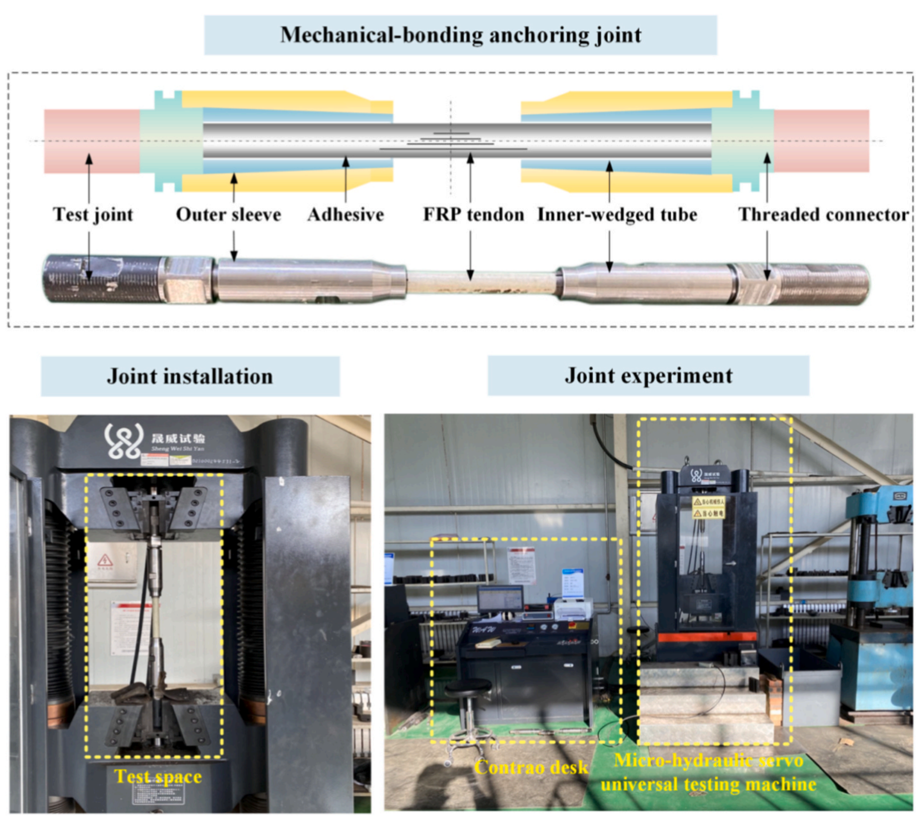


Fig. 13. Morphology of carbon/glass hybrid rod with (a) structure, (b) microstructure and (c) preparation.

relative sliding of the conical surfaces to generate radial compressive force, thereby ensuring the mechanical fastening of the FRP tendon. Simultaneously compress and expel the adhesive to guarantee complete filling of the interface gap, thereby solidifying the anchoring joint. The anchoring joint has been installed at both ends of the FRP tendon according to the given process.

According to GB/T 13096-2008, "Test Methods for Mechanical Properties of Extruded Glass Fiber Reinforced Plastic Rods," apply the controllable micro-hydraulic servo universal testing machine (WAW-1000F) manufactured by Weihai Shengwei Testing Machine Co., Ltd. Use the displacement control at a rate of 5 mm/min to conduct a connection test of the mechanical-bonded wedge anchoring joint as shown in Fig. 13. Persist in loading until either the rod experienced cracking or the joint connection failed. Record the ultimate tensile strength and the failure mode of the anchoring joint, perform five repetitions and obtain the average tensile strength.

5.2. Load-displacement curve

The load-displacement response of the mechanical-bonding anchor joint under tensile displacement is shown in Fig. 14. The experimental data shows that the FRP tendon failed due to splitting at a joint displacement of 7.3 mm, with an ultimate tensile load of 452.4 kN. At the failure point, the slip displacements of the upper and lower joints measured 7.6 mm and 8.0 mm, respectively, while the elongation of the FRP tendon was 1.7 mm. It is important to mention that the gripping ends of the joint did not exhibit any significant damage, which suggests that the mechanical-bonding anchor system has excellent anchoring performance and reliable load transfer.

When tendon failure was ignored, the numerical simulation result indicated a load of 512.3 kN at the same displacement of 7.3 mm. This number is slightly higher than the experimental result (452.4 kN), with a 13.2 % relative error, which indicates that the numerical model can adequately represent the mechanical behavior of the joint before tendon failure occurs. The load continues to increase up to 869.6 kN at a displacement of 14.1 mm, after which a sharp decline occurs due to overall structural instability. The error comes from the simplified boundary conditions and material modelling in the simulation. Perfect alignment and consistent preload were assumed, however actual tests found assembly defects and irregular preload, leading to early damage. Moreover, homogenised material characteristics failed to account for actual composite defects, resulting in a delay of simulated damage which led to an overestimation of load.

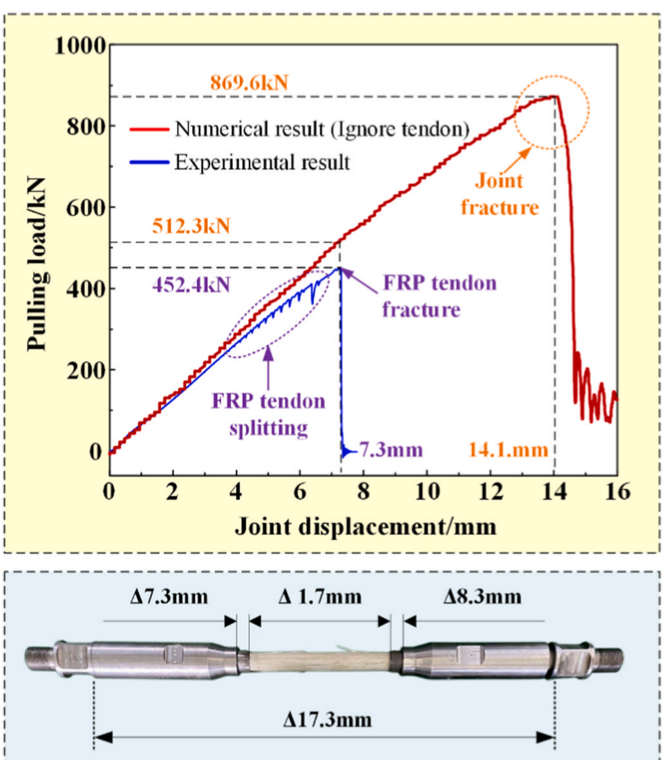


Fig. 14. Load-displacement curve of optimized mechanical-bonded anchor joint.

5.3. Anchoring efficiency

The test results show that the mechanical-bonding anchor joint developed in this study performs successfully shown in Table 6. The primary failure mode was the splitting failure of the FRP tendon. The system achieved an average anchorage efficiency of 95.2 %, with a maximum efficiency of 101.0 %. The appearance of anchoring efficiencies surpassing 100 % is primarily related to the real tensile strength of the FRP tendons exceeding their nominal standard values, which suggests that the anchorage design may completely utilize the true load-bearing potential. These results underscore the necessity of precisely considering the strength and failure criteria of the FRP tendon in the mechanical analysis of mechanical-bonding anchor joints.

6. Conclusion

This research created an innovative mechanical-bonding anchor joint that integrated the mechanical effect of the wedged sleeve with the interface stress optimization provided by the adhesive, resulting in effective anchoring of FRP tendons. It has addressed the problem of early radially crushing failure of FRP tendon, providing a novel approach for achieving efficient and stable anchorage. The main conclusions are as follows.

(1) A three-dimensional finite element model using the Johnson-Cook constitutive model and the cohesive zone model was developed to characterize the mechanical response, stress distribution, and progressive failure process of the joint subjected to tensile displacement. The predicted load value at a displacement of 7.3 mm (512.3 kN) exhibited a relative error of 13.2 % in relation to the actual value (452.4 kN), hence confirming the accuracy of the simulation model.

(2) The critical parameters affecting the mechanical-bonding anchoring system are disclosed: at an differential angle of approximately 0.5°, the pull-out load reached its peak, although the failure mode transitioned to brittle; as the differential angle rises to 2°, both the load and compressive stress decreased by roughly 61 %. Increasing the wedge angle (from 0.9° to 1.7°) can improve the bearing capacity (by 15.2 %), although it would elevate the compressive stress (by 14 %). Extending the joint length (from 120 mm to 180 mm) and the slotting length (from 90 mm to 110 mm) can enhance the load capacity (by 10.6 % and 13.8 % respectively) while markedly diminishing the compressive stress (by 15.7 % and 27.2 % respectively) and optimizing the uniformity of stress distribution.

Table 6
Anchoring efficiency of mechanical-bonding anchor joint.

Number	Tensile strength/MPa	Anchoring efficiency/%
1	1108.8	88.7
2	1136.9	90.9
3	1192.4	95.4
4	1250.1	100.1
5	1262.2	101.0

(3) A highly precise predictive model was developed with the GA-BP neural network. The R values for the training sets predicting pulling load and compressive stress were 0.949 and 0.842, respectively. The largest prediction errors for the verification samples were 12.2 % and 16.5 %, respectively. The NSGA-II algorithm for multi-objective optimization yielded the ideal parameter combination: chord angle 1.2°, differential angle 0.8°, joint length 150 mm, and inner tube slotting length 125 mm.
(4) The testing results indicate that the improved joint attained an average anchoring efficiency of 95.2 %, with a maximum efficiency of 101.0 %. This markedly surpasses the performance of the unoptimized mechanical anchoring, which failed when the rod body was crushed under a force of 300 kN, exhibiting an efficiency of merely 63.1 %.

CRediT authorship contribution statement

Yanwen Zhang: Writing – original draft. **Shuqing Wang:** Software, Funding acquisition. **Mingqiang Xu:** Writing – review & editing, Conceptualization. **Wencheng Liu:** Validation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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